

AP CALCULUS AB – UNIT 3

3.1 The Chain Rule



Name: _____ Class: _____ Date: _____

Lesson Overview and Learning Objectives

- Recognize composite functions and identify their inner and outer functions.
- Differentiate composite functions using the Chain Rule in prime and Leibniz notation.
- Combine the Chain Rule with power, trigonometric, exponential, logarithmic, product, and quotient rules.
- Use tables, graphs, and contextual information to evaluate derivatives of compositions.

Keys: Essential Knowledge and Vocabulary

Composite function	A function of the form $f(g(x))$; the output of g becomes the input of f .
Chain Rule	$\frac{d}{dx}f(g(x)) = f'(g(x))g'(x).$
Leibniz form	If $y = f(u)$ and $u = g(x)$, then $\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx}$.
Nested composition	Differentiate from the outside inward and multiply by every inner derivative.
Evaluation from data	$(f \circ g)'(a) = f'(g(a))g'(a)$; locate $g(a)$ first.
Units	Rates multiply so intermediate units cancel, leaving output units per input unit.

Explanation, Strategy, and Common Errors

- Mark the innermost expression before differentiating. Rewrite radicals and reciprocals as powers when useful.
- Differentiate the outside while leaving the inside unchanged, then multiply by the derivative of the inside.
- For several layers, repeat the process until the derivative reaches the original independent variable.
- When product or quotient structure is present, preserve that structure and apply the Chain Rule inside its factors.
- Do not write $f'(x)g'(x)$ for the derivative of $f(g(x))$; the outer derivative is evaluated at $g(x)$.

Solved Example: Power outside

Differentiate $y = (3x^2 - 5x + 1)^4$.

Answer and Reasoning

Let $u = 3x^2 - 5x + 1$. Then $\frac{dy}{dx} = 4u^3 \frac{du}{dx} = 4(3x^2 - 5x + 1)^3(6x - 5)$.

Solved Example: Three nested layers

Differentiate $y = e^{\sin(2x)}$.

Answer and Reasoning

Differentiate exponential, then sine, then $2x$: $y' = 2e^{\sin(2x)} \cos(2x)$.

Solved Example: Product and Chain Rules

Differentiate $y = x^2 \ln(1 + x^3)$.

Answer and Reasoning

$$y' = 2x \ln(1 + x^3) + x^2 \frac{3x^2}{1 + x^3} = 2x \ln(1 + x^3) + \frac{3x^4}{1 + x^3}.$$

Shared Representation / Data

x	$f(x)$	$f'(x)$	$g(x)$	$g'(x)$
1	4	-2	2	3
2	-1	5	3	-1
3	6	4	1	2

AP Exam Language

For every derivative in context, include the sign, units, input value, and a complete verbal interpretation. For every existence claim, state the relevant hypothesis or one-sided comparison.

Leveled Practice**Easy Foundations**

Build fluency with the definition, notation, and one-step applications.

1. Identify inner and outer functions for $y = \sqrt{5x^2 + 1}$.

2. Differentiate $y = (2x - 7)^5$.

3. Differentiate $y = \sin(4x)$.

4. Differentiate $y = \ln(3x + 2)$.

Medium Applications

Connect representations and apply the lesson procedure in familiar settings.

5. Differentiate $y = e^{x^2-3x}$.

6. Differentiate $y = (1 + \tan x)^6$.

7. For $y = \sqrt{1+x^4}$, find $y'(1)$.

8. If $h = f \circ g$, $g(2) = 3$, $g'(2) = -1$, and $f'(3) = 5$, find $h'(2)$.

Hard Reasoning

Select a method, justify conclusions, and combine several ideas.

9. Differentiate $y = x^3 \cos(x^2)$.

10. Differentiate and simplify $y = \ln((x^2 + 1)^3)$.

11. Write the tangent line to $y = (x^2 + 3)^{-1}$ at $x = 1$.

12. Using the table, find $(f \circ g)'(2)$ and decide whether $(g \circ f)'(2)$ is determined.

Difficult Analysis

Work through less-direct algebra, subtle hypotheses, and multi-step reasoning.

13. Find all inputs where the derivative of $y = e^{x^2-4x}$ is zero.

14. For $f(x) = \sin(x^2)$, find $f''(0)$.

15. Find a so $F(x) = (ax + 1)^4$ has $F'(0) = 8$.

Challenge / AP Extension

Synthesize the topic in unfamiliar, proof-oriented, or parameter-based problems.

16. If differentiable f is even, prove that f' is odd.

17. Let $H(x) = g(g(x))$. If $g(1) = 2$, $g'(1) = -3$, and $g'(2) = 4$, find $H'(1)$.

AP-Style Multiple-Choice Practice

Part A: No Calculator

Choose the best answer. Justification is not required on the AP multiple-choice section, but show reasoning while practicing.

1. If $y = (1 - 3x^2)^5$, then y' is

- (A) $5(1 - 3x^2)^4$ (B) $-30x(1 - 3x^2)^4$
(C) $-6x(1 - 3x^2)^5$ (D) $5(-6x)^4$

2. If $h = f \circ g$, which equals $h'(a)$?

- (A) $f'(a)g'(a)$ (B) $f'(g(a)) + g'(a)$
(C) $f'(g(a))g'(a)$ (D) $f(g'(a))$

3. Using the table, $(f \circ g)'(2)$ equals

- (A) -5 (B) -4
(C) 4 (D) 5

4. A student writes $\frac{d}{dx} \ln(x^2 + 1) = \frac{1}{x^2 + 1}$. The missing factor is

- (A) x^2 (B) $2x$
(C) $x^2 + 1$ (D) $\ln x$

Part B: Graphing Calculator Permitted

Use a calculator only for numerical evaluation, graphing, or regression as indicated. Preserve mathematical setup.

5. At $x = 0.8$, the derivative of $F(x) = e^{\sin(x^2)}$ is closest to

- (A) 1.36 (B) 1.88
(C) 2.33 (D) 3.41

6. At $x = 1.3$, the slope of $y = \ln(1 + x^4)$ is closest to

- (A) 1.29 (B) 1.87
(C) 2.28 (D) 2.94

AP-Style Free-Response Practice

Calculator Permitted

A quantity is modeled by $P(t) = 500e^{-0.2(t^2+1)}$, where t is measured in hours.

(a) Find $P'(t)$. [2 points]

(b) Estimate $P'(1.5)$. [2 points]

(c) Interpret the result. [2 points]

No Calculator

Functions satisfy $f(2) = 3$, $f'(2) = -4$, $g(1) = 2$, $g'(1) = 5$, and $g'(3) = -2$. Let $H(x) = g(f(g(x)))$.

(a) Write $H'(x)$. [2 points]

(b) Find $H'(1)$. [3 points]

(c) Explain why the first factor cannot be replaced by $g'(x)$. [2 points]